QSCP Hamilton Inspection Department

Inspection Report

Site: Hamilton Plant

No.	Status	Date
QSCP-INS-2015-024	Issued	2015-05-04

External Visual Inspection and CUI Windows - ATU-100

Scope and Objectives

This report details the 5-yearly external visual inspection and CUI window examination for the ATU-100. The scope follows the requirements for insulated lines operating within the CUI range as defined in QSCP-IP-ATU100 Rev 2. Douglas Ferreira served as the API 570 inspector responsible for the assessment of piping integrity and the verification of insulation condition across the unit.

The inspection focuses on identifying external degradation, coating failure, and moisture ingress that could lead to localized corrosion. All activities were performed in accordance with the practices outlined in API 570 and API 574 to ensure compliance with site asset integrity standards. The primary objective is to validate the current condition of the piping systems against the findings recorded in QSCP-INS-2010-021.

Responsibilities

Douglas Ferreira, acting as the Inspection Supervisor for the QSCP Hamilton Inspection Department, oversaw the execution of the CUI program and the visual assessment of all accessible piping circuits. The inspector is responsible for determining if any observed wall loss requires the issuance of an NCR based on the established t-min for each circuit.

The QSCP Hamilton Inspection Department is tasked with maintaining the inspection history and updating the remaining life calculations. Per QSCP-IP-ATU100 Rev 2, the department must ensure that profile RT or insulation windows are executed every 5 years for lines identified as susceptible to CUI. This ensures that the short-term and long-term corrosion rates are accurately monitored.

Review of Previous Inspection Data

A review of QSCP-INS-2010-021 indicates that the previous 5-yearly external visual inspection identified damaged jacketing and wet insulation on lines 4-RA-104 and 4-LA-104. These specific locations were flagged for mandatory CUI inspection during this cycle to determine if significant external corrosion has occurred under the compromised thermal protection.

The findings from 2010-05-10 served as the baseline for the current examination. Following the recommendations in QSCP-INS-2010-021, the inspection team prioritized the removal of insulation at the previously identified wet spots to perform direct visual examination of the pipe surface, adhering to the guidelines in API 574.

CUI Program Execution

In accordance with QSCP-IP-ATU100 Rev 2, insulation windows were cut at designated locations on the ATU-100 piping. This 5-yearly requirement is designed to detect corrosion that is not visible during standard external walks. The inspection targeted the rich amine and regenerator overhead circuits, which are subject to specific thickness interval constraints under site rules.

The examination of the pipe surface through these windows allows for the verification of the coating integrity. Douglas Ferreira confirmed that the inspection locations selected represent the highest risk areas for moisture collection, consistent with the methodology described in API 570 for managing CUI in insulated carbon steel and stainless steel systems.

External Visual Findings

The 5-yearly external visual inspection covered all accessible piping within the unit. The general condition of the pipe supports, hangers, and small-bore connections was assessed. Observations were compared against the criteria in API 574 to identify any mechanical damage or atmospheric corrosion that could impact the remaining life of the assets.

Specific attention was paid to the areas where QSCP-INS-2010-021 had previously noted insulation deficiencies. The results of the CUI window inspections and the external visual survey provide the necessary data to calculate the current corrosion rates and schedule the next thickness readings within the class maximum intervals.

Sign-off	Name	Function	Date	Signature
Prepared	Douglas Ferreira	Inspection Supervisor, API 570 inspector	2015-05-04	_____